AutoML Experiment Report: Retina_Blood_Vessel_for_segmentation

AutoHealth

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Abstract

This report details an automated machine learning (AutoML) experiment for retinal blood vessel segmentation from fundus images. The task is a binary medical image segmentation problem, aiming to produce accurate pixel-level vessel masks. The dataset comprised 100 images (512x512 pixels) split into training, validation, and test sets. We employed a deep ensemble strategy combining two U-Net models with EfficientNet-B0 and ResNet-18 backbones, enhanced with Monte Carlo Dropout for uncertainty estimation. Key findings include a final test Dice coefficient of 0.684 ± 0.030, indicating moderate segmentation performance with room for improvement. Uncertainty analysis revealed a weak correlation (0.303) between model uncertainty and prediction error. Temperature scaling calibration successfully reduced the Expected Calibration Error (ECE) from 0.146 to 0.112. The analysis highlights challenges related to class imbalance, limited dataset size, and the need for more robust feature learning. Recommendations for future iterations focus on advanced data augmentation, architectural refinements, and leveraging color channel information to improve model accuracy and reliability.

1 Data Analysis

1.1 Dataset Overview and Modality

The experiment utilized a dataset of retinal fundus images for binary vessel segmentation. The dataset was organized into standard splits: a training set (72 images), a validation set (8 images), and a test set (20 images), totaling 100 paired RGB images and corresponding grayscale mask files. All images and masks were in PNG format with a uniform resolution of 512x512 pixels. The dataset structure facilitated straightforward loading, with corresponding image and mask files sharing identical filenames in separate 'image/' and 'mask/' directories for each split.

1.2 Target Distribution and Class Imbalance

A critical characteristic of the data is significant class imbalance. The vessel pixels (foreground) constitute only approximately 12.3% of all pixels on average across all splits, while background pixels dominate. This imbalance is consistent, with the test set showing a vessel pixel ratio of 0.087. Such imbalance poses a challenge for segmentation models, which may become biased towards predicting the majority background class if not properly addressed through loss function design or sampling strategies.

1.3 Data Quality and Mask Characteristics

Data quality was generally good with no corrupted files. The segmentation masks correspond to a binary vessel extraction task, with vessel regions annotated against the background. During preprocessing and evaluation, masks were handled consistently with a binary segmentation setting, using a fixed threshold to produce the final vessel map where needed.

1.4 Feature Analysis and Performance Clues

Analysis of the relationship between image features and the target mask revealed valuable insights. The most significant distinguishing factor between vessel and background pixels was color, particularly in the red (R) channel. The average pixel value difference between vessel and background regions was largest

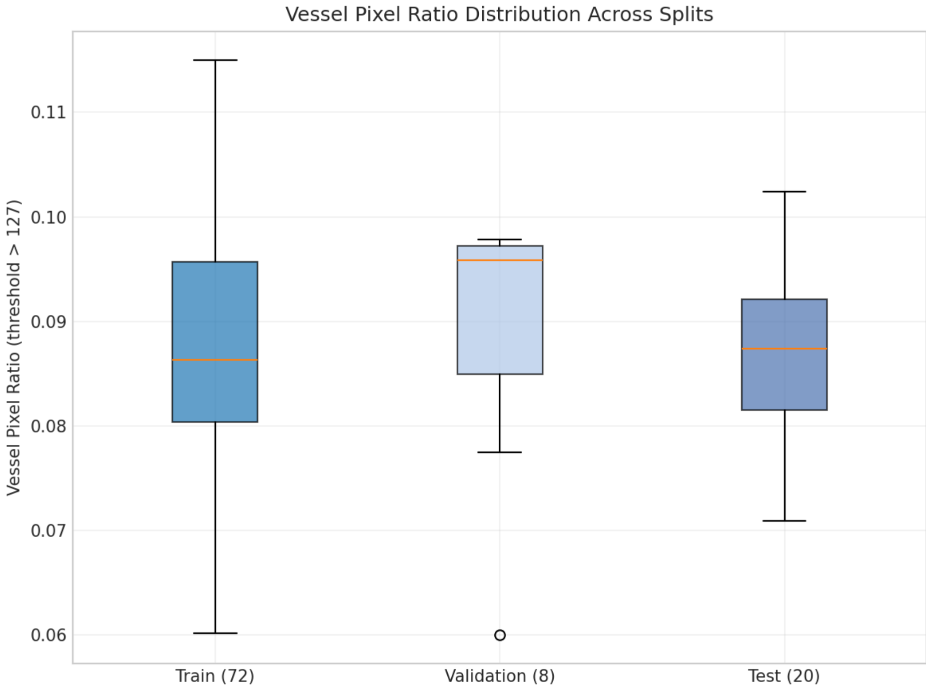

Figure 1: Distribution of vessel pixel ratio across dataset splits. The boxplot shows the median, quartiles, and range of the proportion of vessel pixels per image in the training, validation, and test sets. The consistent low ratio confirms significant class imbalance. The wider distribution in the training set suggests greater variability, which may help learning but also increases overfitting risk due to the small sample size.

in the R channel ($\Delta = 63.6$), compared to the green ($\Delta = 28.7$) and blue ($\Delta = 21.6$) channels. This suggests that models could benefit from emphasizing or learning features from the red channel.

Other notable findings include a slight systematic offset of the vascular structure from the image center and minor differences in image sharpness between the validation set and the other splits, which could subtly affect model evaluation.

2 Model Training

2.1 Data Preprocessing

The preprocessing pipeline was designed to handle the specific characteristics of the dataset while augmenting the limited training data. Images (RGB) and masks (grayscale) were normalized by scaling pixel values to the range [0, 1]. Masks were treated as soft labels during training (values in [0,1]) and binarized with a 0.5 threshold only for final evaluation. To combat overfitting from the small dataset (80 images after merging train and original val sets), a set of lightweight augmentations was applied only during training using the Albumentations library. These included horizontal/vertical flips (p=0.5), random rotation within $\pm 15°$ (p=0.5), random brightness/contrast adjustments (limit=0.05, p=0.5), and additive Gaussian noise (p=0.3). All spatial transformations were applied synchronously to images and masks.

2.2 Model Architecture

We employed a Deep Ensemble approach for robust performance and uncertainty estimation. The ensemble consisted of two independent U-Net models with different encoder backbones initialized with ImageNet pre-trained weights:

- **Model A**: U-Net with an EfficientNet-B0 encoder.
- **Model B**: U-Net with a ResNet-18 encoder.

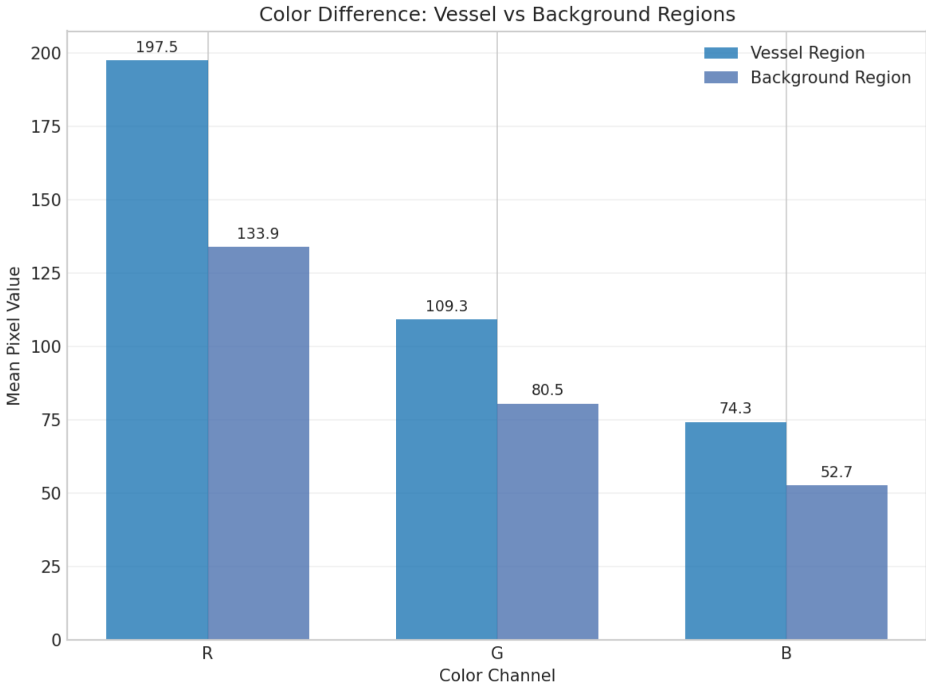

Figure 2: Average RGB pixel values for vessel vs. background regions in a sample image. The bar chart clearly shows that the red channel provides the greatest contrast between vessels and background, making it the most informative single channel for this segmentation task. This finding can guide model architecture choices or input preprocessing.

Both models had standard U-Net decoder pathways. To estimate *aleatoric* uncertainty, dropout layers (p=0.1) were inserted after each convolutional layer in the decoder and kept active during both training and inference. The ensemble's *epistemic* uncertainty was derived from the disagreement between the two models' predictions.

2.3 Hyperparameters

Key hyperparameters were chosen to facilitate stable training given the data constraints:

- **Loss Function**: A combination of Dice Loss and Binary Cross-Entropy (BCE) Loss. The BCE loss used a dynamically calculated 'pos_weight' per training batch to adapt to the varying vessel ratio in each batch, directly addressing class imbalance.
- **Optimizer**: AdamW with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4}.
- **Learning Rate Schedule**: A warm-up cosine annealing scheduler. The learning rate linearly increased from 1×10^{-6} to 1×10^{-4} over 5 epochs, then followed a cosine decay to 1×10^{-6} over 25 epochs.
- **Training Duration**: Maximum of 50 epochs with early stopping (patience=10) monitoring the validation loss.
- **Batch Size**: 16.

2.4 Training Process

The training process revealed differences in the learning dynamics of the two ensemble members. Model B (ResNet-18) achieved a lower best validation loss (1.007) compared to Model A (EfficientNet-B0, validation loss 1.211). Model A triggered early stopping at epoch 25, while Model B trained for the full 50 epochs. The training history shows that Model B's validation loss decreased more consistently.

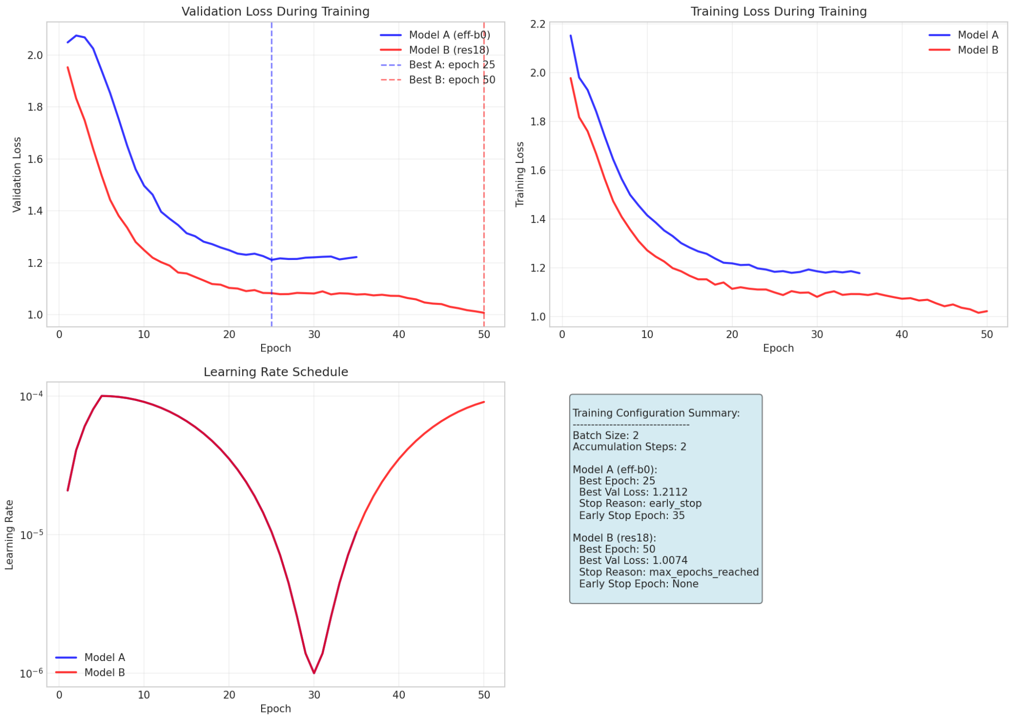

Figure 3: Complete training history for both ensemble models. The left panels show the training and validation loss curves for Model A (eff-b0) and Model B (res18). Model B achieves a lower final validation loss. The right panel shows the learning rate schedule. The divergence in performance suggests the ResNet-18 backbone may be more suitable for this specific task and dataset size.

2.5 Model Performance

The primary evaluation metric was the Dice Similarity Coefficient (Dice). On the held-out test set of 20 images, the ensemble model achieved a mean Dice score of **0.684** with a standard deviation of 0.030. Individual image scores ranged from 0.624 to 0.743.

Table 1: Test Set Performance Summary

Metric	Value
Mean Dice Coefficient	0.684
Dice Std. Deviation	0.030
Dice Range	[0.624, 0.743]
Brier Score	0.0587

This level of performance indicates the model successfully learns the segmentation task but has room for improvement, particularly in consistently segmenting finer vessel structures across all test cases. The variation in scores suggests sensitivity to image-specific characteristics.

3 Uncertainty Analysis

A comprehensive uncertainty framework was implemented, combining epistemic (model) uncertainty from the ensemble and aleatoric (data) uncertainty from MC Dropout. The total uncertainty per pixel was visualized and analyzed.

3.1 Uncertainty Visualization and Interpretation

Uncertainty heatmaps provide a clinician-friendly tool to identify regions where the model's prediction is less reliable. These areas often correspond to thin vessel boundaries, vessel branch points, or regions with low contrast.

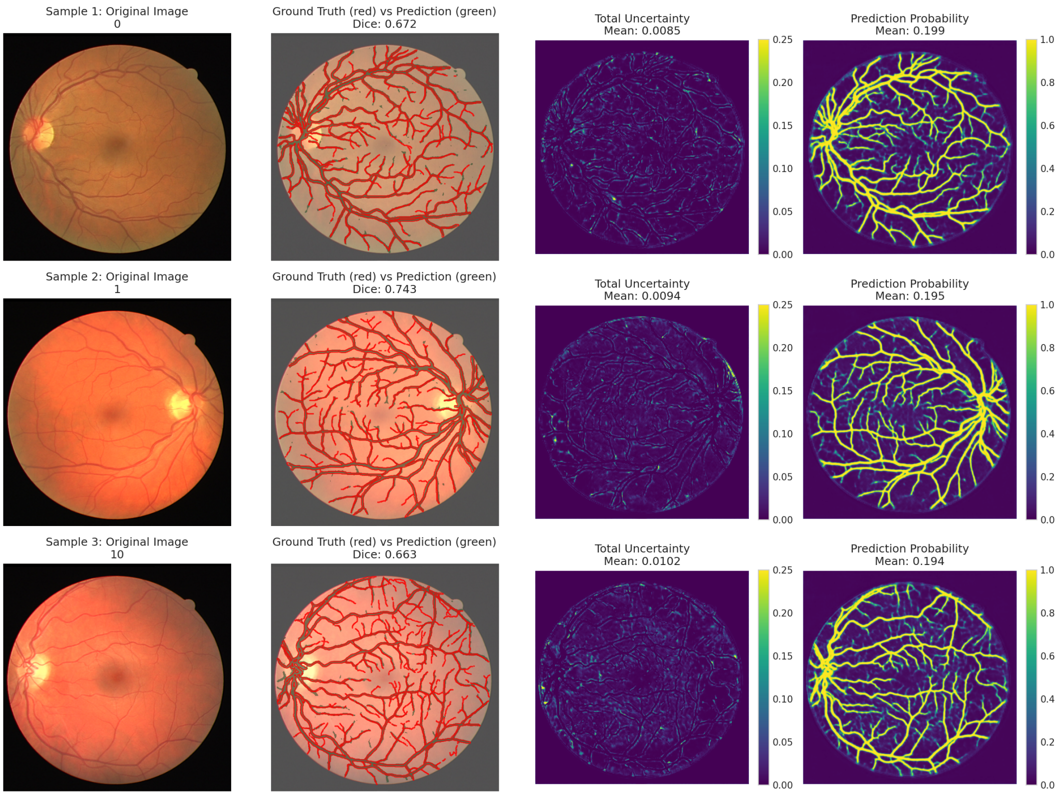

Figure 4: Uncertainty visualization for three test samples. For each sample: Row 1 shows the original fundus image. Row 2 overlays the ground truth mask (red outline) and the model's binary prediction (green area), with the Dice score. Row 3 shows the total uncertainty heatmap (brighter color indicates higher uncertainty). High uncertainty often aligns with complex vascular junctions or faint vessels.

3.2 Quantitative Uncertainty Metrics

The correlation between the total uncertainty and the absolute prediction error was calculated to be 0.303, indicating a weak positive relationship. This means that, on average, pixels with higher model uncertainty are slightly more likely to be misclassified, but the relationship is not strong enough for uncertainty to be a perfect error predictor.

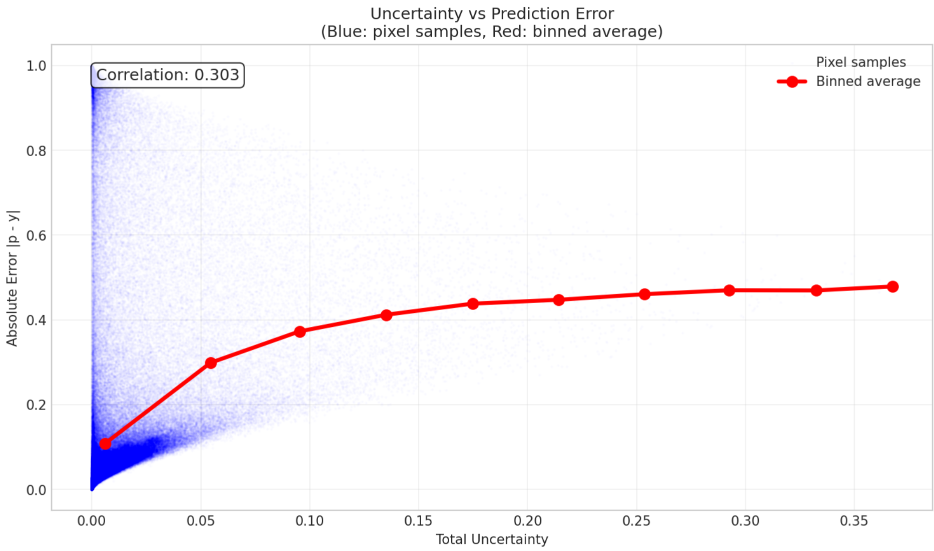

Figure 5: Scatter plot and binned average trend (red line) showing the relationship between pixel-wise total uncertainty and absolute prediction error. The weak positive correlation (0.303) suggests that while higher uncertainty sometimes indicates higher error, there are many cases of high uncertainty with low error and vice-versa.

3.3 Model Calibration

The model's probabilistic outputs were calibrated using Temperature Scaling on a held-out calibration set. This process adjusted the confidence of the predictions to better match their empirical accuracy. The Expected Calibration Error (ECE) improved from 0.146 to 0.112 after calibration, meaning the model's predicted probabilities became more reliable indicators of its true correctness likelihood.

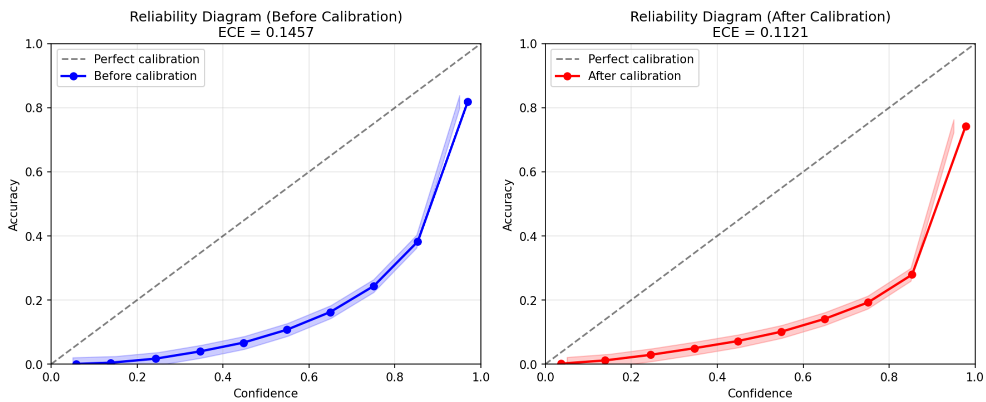

Figure 6: Reliability diagrams before (left) and after (right) temperature scaling calibration. The dashed line represents perfect calibration. The reduction in Expected Calibration Error (ECE) from 0.1457 to 0.1121 shows that calibration successfully made the model's confidence scores more truthful.

3.4 Using Uncertainty for Reject Option

A key clinical application of uncertainty is the "reject option": abstaining from making a prediction in high-uncertainty regions, potentially flagging them for expert review. Analysis shows that applying an uncertainty threshold to filter out the most uncertain pixels leads to a trade-off: higher Dice scores on the remaining pixels but at the cost of discarding an increasing portion of the image.

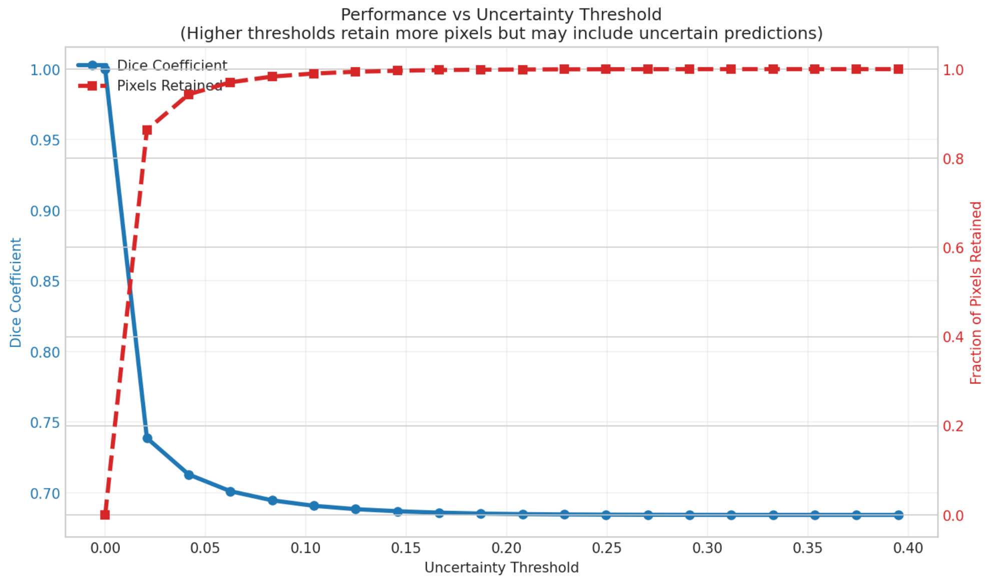

Figure 7: Performance trade-off when using an uncertainty threshold. The left y-axis (blue line) shows the Dice coefficient calculated only on pixels with uncertainty below the threshold. The right y-axis (orange line) shows the proportion of pixels retained. As the threshold increases (discarding fewer pixels), the Dice score quickly drops and stabilizes, indicating that a small set of very uncertain pixels disproportionately affects performance.

4 Conclusion

This AutoML experiment developed a deep ensemble model for retinal blood vessel segmentation, achieving a moderate Dice score of 0.684. The integrated uncertainty quantification framework provides valuable insight into model confidence, with calibration improving reliability. The analysis identified key factors influencing performance: significant class imbalance, the small overall dataset size, and the high informative value of the red color channel.

Actionable Guidance for Clinicians and Developers:

1. **Model Output**: The segmentation masks should be used with an understanding of their moderate accuracy. They are suitable for initial screening or quantitative analysis where approximate vessel area is needed, but critical clinical decisions should not rely solely on this output.

2. **Uncertainty Maps**: The provided uncertainty heatmaps are a crucial tool. Regions of high uncertainty should be reviewed carefully by a human expert, as they often indicate challenging anatomical structures or potential model errors.

3. **Path Forward**: To improve performance, future work should prioritize: 1) Acquiring or synthesizing more training data, 2) Implementing more aggressive, task-specific data augmentations (e.g., vascular structure deformations), 3) Experimenting with loss functions more robust to extreme imbalance (e.g., Tversky, Focal Loss), and 4) Explicitly leveraging the red-channel dominance through dedicated model pathways or preprocessing.

The experiment demonstrates a functional pipeline for automated segmentation with built-in confidence assessment, establishing a foundation for more robust and clinically trustworthy tools in retinal image analysis.